

High-Capacity Aqueous Potassium-Ion Batteries for Large-Scale Energy Storage

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Large-scale energy storage is a key strategy to effectively utilize intermittent renewable energy.^[1] The electrochemical storage of energy using rechargeable batteries plays an important role in this context. Despite the many types of rechargeable batteries that have been investigated, few can meet the requirements of capacity, stability, cost, and safety simultaneously. Metal hydride/nickel cells suffer from the low energy density, redox-flow batteries lack high-rate performance, and lithium-ion batteries are too expensive and flammable electrolyte can also cause safety issues.^[2] Thus, the development of affordable and safe electric storage batteries for large-scale applications has become a priority.

Aqueous electrolyte rechargeable batteries based on alkali ion intercalation are attractive candidates because of their inherent safety and low cost.^[1,3] Aqueous lithium-ion batteries have been comprehensively studied,^[4] but they generally show poor stability (the capacity retention is typically less than 50% after 100 cycles).^[5] Furthermore, there is a relatively low natural abundance of lithium. Aqueous sodium-ion batteries meet the low-cost requirement due to the high abundance of sodium^[6] but suffer from low-capacity issues.^[7] Potassium is also a naturally abundant element and so aqueous rechargeable batteries based on the K^+ shuttle ion are a potentially viable technology. Breakthroughs in cathode^[8] and anode^[9] materials have demonstrated the potential of potassium-ion batteries. They can be much cheaper than aqueous lithium-ion batteries and can exhibit superior capacities in comparison to aqueous sodium-ion batteries. Nevertheless, the higher ionization potential and larger ionic radius of potassium has so far limited the choice of electrode materials for potassium-ion batteries.

One promising class of electrode material for potassium-ion batteries is the Prussian Blue analogues (PBAs).^[10] PBAs have the perovskite-type structure with the chemical formula of $A_xM[Fe(CN)_6]_y \cdot mH_2O$ (where A = alkali metal, M = transition metal, and $0 \leq x \leq 2$, $y < 1$), in which the nitrogen-coordinated transition metal cations and carbon-coordinated Fe sites form an open framework bridged by cyanide groups ($-C \equiv N-$).

Within the open framework, large interstitial sites can host alkali metals and form a 3D channel for fast transport of metal ions.^[11] Recently developed aqueous potassium-ion batteries based on PBAs^[12] have shown reasonable performance although they exhibited low capacity due to the low potassium-ion content within the PBA.^[12a-c] Many reported PBAs can supply only one electron per formula unit due to the higher oxidation state of the transition metal in the interstitial sites (e.g., $Na_{1.4}Cu^{II}_{1.3}Fe^{II}(CN)_6$,^[13] $Na_{1.94}Ni^{II}_{1.03}Fe^{II}(CN)_6$,^[14] and $Na_2Zn^{II}_3[Fe^{II}(CN)_6]_2 \cdot 9H_2O$).^[15] Thus, the capacity, which relies on the one-electron $Fe^{III/II}$ process, is below 70 mA h g^{-1} . However, some PBAs are capable of storing up to two alkali metal ions per formula unit and undergo multi-electron processes. PBAs containing manganese or cobalt have been reported to deliver more than 100 mA h g^{-1} reversible capacity in lithium-ion and sodium-ion batteries,^[16] although their cycling performances were poor. Iron (II) hexacyanoferrate compounds (sometimes referred to as Prussian White or Everitt's salt) exhibit two one-electron redox processes and have been prepared with high sodium content.^[17] Batteries using this material exhibited high specific capacity and good capacity retention (90% after 200 cycles) for the aqueous sodium-ion batteries. Rhombohedral Prussian White with sodium cations ($Na_{1.92}Fe[Fe(CN)_6]$) in non-aqueous sodium-ion batteries also demonstrated good performance.^[18] To date, the use of potassium-rich iron hexacyanoferrate cathode materials in aqueous electrolyte batteries has not been reported.

Here, we report the synthesis of high-potassium-ion-content potassium iron (II) hexacyanoferrate dehydrate nanocubes and their performance as electrodes in aqueous electrolyte potassium-ion batteries. The potassium-rich iron (II) hexacyanoferrate can supply two electrons per formula unit and therefore deliver exceptionally high capacities in aqueous potassium-ion batteries (up to 120 mA h g^{-1} , the highest value for the aqueous potassium-ion batteries to date). As revealed by electrochemical analysis, ex situ X-ray diffraction (XRD), ex situ Fourier transform infrared (FTIR) spectroscopy, ex situ electron energy-loss spectroscopy (EELS) and kinetics analyses, the potassium iron (II) hexacyanoferrate dihydrate nanocubes maintain a low-strain open framework structure during the charge and discharge processes where potassium ions are inserted/extracted. These processes occur with fast kinetics and are highly reversible as identified by electrochemical measurements and density functional theory (DFT) calculations. Therefore, we present for the first time an aqueous potassium-ion battery with exceptional capacity that is made from safe and cost effective materials. The developed battery system can deliver on each of the key aspects required for genuine large-scale energy-storage devices.

During the synthesis of PBAs in aqueous solution, the nucleation and growth phase is generally a rapid process, which

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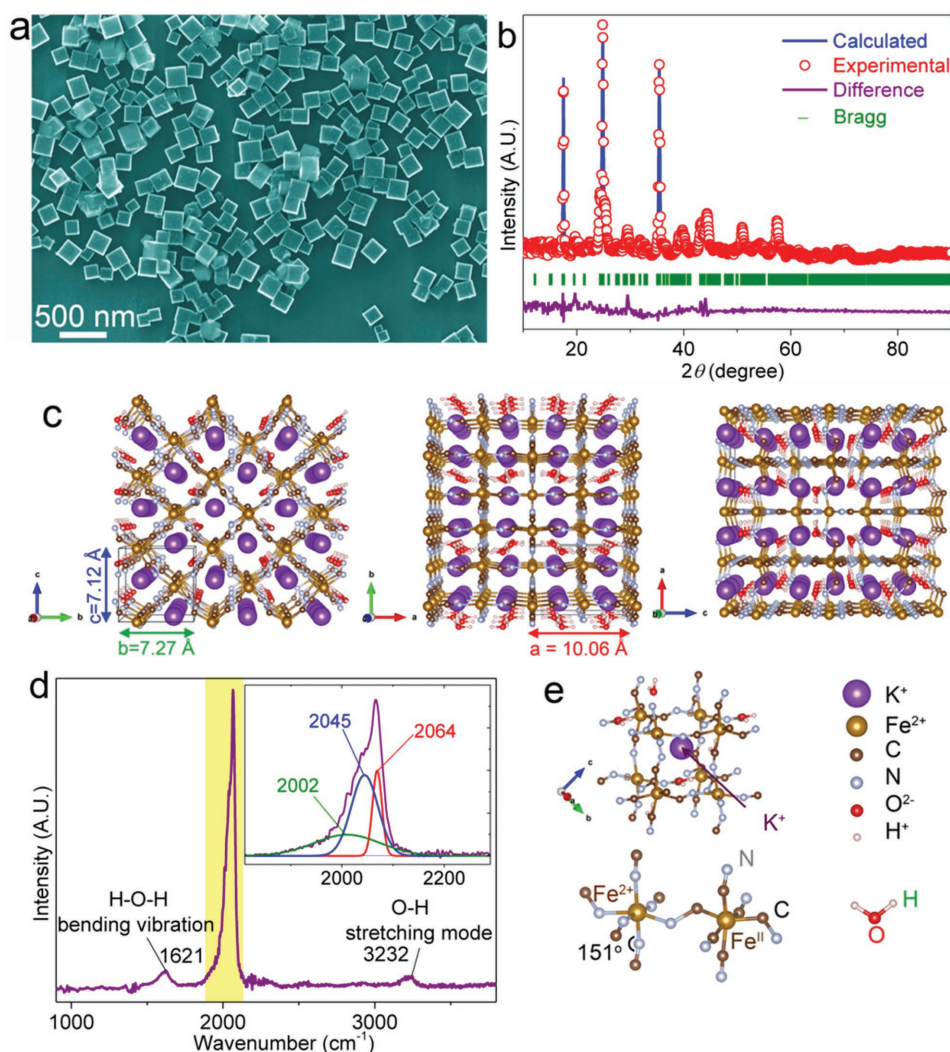


Figure 1. a) FESEM image of potassium iron hexacyanoferrate dihydrate particles. b) Rietveld refinement pattern of X-ray diffraction data for potassium iron hexacyanoferrate dihydrate particles. The observed and calculated intensities are represented by the red circles and the blue solid line, respectively. The bottom purple-colored line shows the fitting residual difference. The Bragg positions are represented by light-green ticks. (The crystal structure was refined to Pmn21 space group with the convergence factors of $\chi^2 = 8.33$, $R_{wp} = 4.35\%$, $R_p = 5.18\%$. The lattice parameters are $a = 10.06$ Å, $b = 7.27$ Å, and $c = 7.12$ Å). c) Refined crystal structure views along different axes. d) FTIR spectrum. Inset is enlarged image. e) Unit formula of potassium iron hexacyanoferrate dihydrate.

makes control of particle size difficult. To maintain a uniform size distribution, surfactants such as poly(vinylpyrrolidone)^[19] have been used. Here, we found that changing the solvent from water to ethylene glycol, which is also a mild reducing agent, gave uniform particle size and maintained the iron in the divalent state. Uniform potassium iron hexacyanoferrate dihydrate nanocubes with an edge length of ≈ 200 nm were obtained using a solvothermal method at 80 °C (Figure 1a and Figure S1, Supporting Information). For comparison, when water was used as a solvent, potassium iron hexacyanoferrate dihydrate with a large particle size distribution was obtained (Figure S2–S4, Supporting Information).

Rietveld-refined XRD was used to identify the crystal structure of the as-prepared potassium iron hexacyanoferrate (Figure 1b). The results show that the particles have an orthorhombic symmetry structure with the Pmn21 space group

(convergence factors: $\chi^2 = 8.33$, $R_{wp} = 4.35\%$, $R_p = 5.18\%$) and have lattice parameters of $a = 10.06$ Å, $b = 7.27$ Å, and $c = 7.12$ Å, rather than the more frequently reported cubic structure.^[20] The corresponding refined structural models along different directions are shown in Figure 1c, in which there are two different iron cation sites forming a double perovskite framework. One iron atom occupies the nitrogen-coordinated sites, forming $\text{Fe}^{\text{II}}\text{N}_6$ octahedra, while the other occupies the carbon-coordinated sites, forming $\text{Fe}^{\text{I}}\text{C}_6$ octahedra (Figure 1e). A 3D framework is formed with bridging $-\text{C}\equiv\text{N}-$ anions to give large interstitial sites ($5.09 \text{ Å} \times 5.19 \text{ Å}$). These sites allow facile insertion/extraction of potassium ions along the [100] direction. Interstitial sites also contain H_2O , which is confirmed by the FTIR spectrum (Figure 1d)^[21] in which the peaks at 3232 and 1621 cm^{-1} are due to the O–H stretching mode and H–O–H bending modes of H_2O respectively. A strong

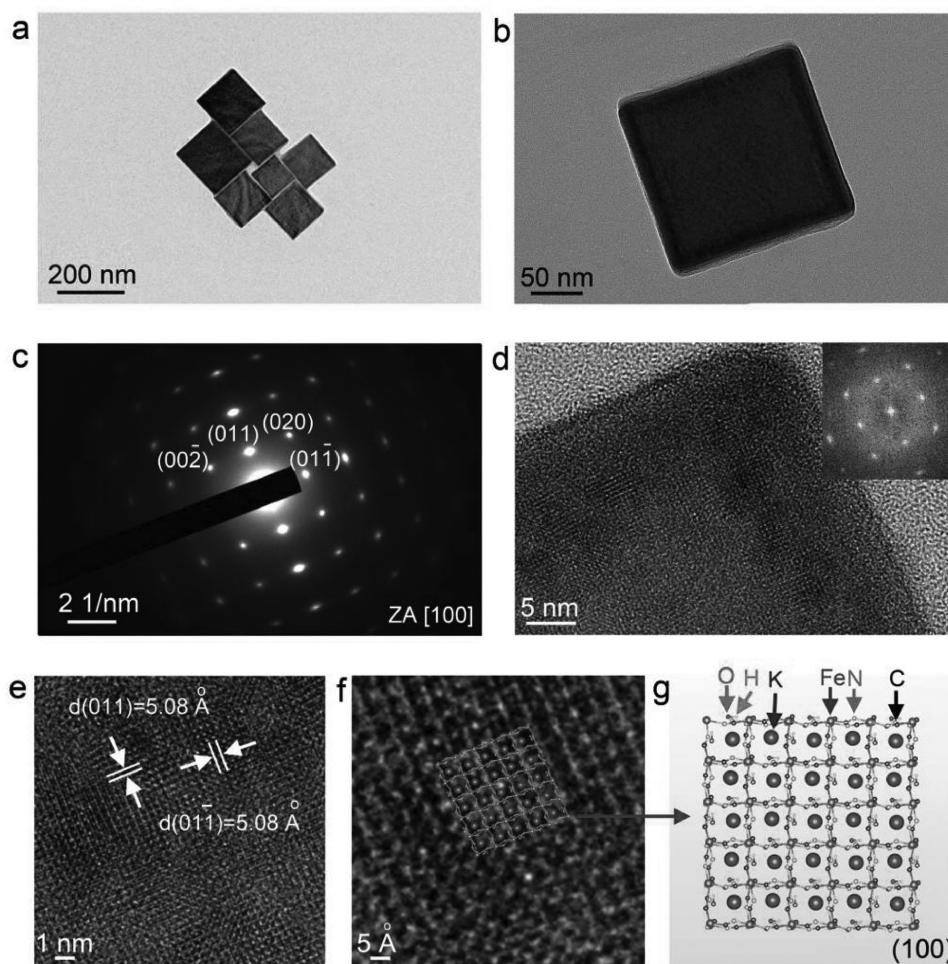


Figure 2. a,b) Low-magnification (a) and medium-magnification (b) TEM images of freestanding particles. c) The SAED patterns of single potassium iron hexacyanoferrate dihydrate particles. d,e) High-magnification TEM images from which the lattice of crystal planes can be observed. The inset in (d) is the FFT patterns. f) HAABF-STEM image, taken from (e), which shows the (100) crystal planes. g) The corresponding atomic models of (100) facets.

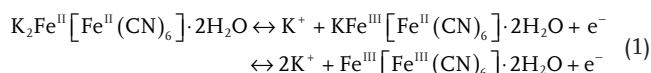
broad band at $\approx 2060\text{ cm}^{-1}$ was fitted with three bands (at 2064, 2045, and 2002 cm^{-1} respectively, inset in Figure 1d), which are assigned to the stretching vibrations of the $\text{C}\equiv\text{N}$ triple bond of cyanide-coordinated Fe^{II} , in agreement with data for related cyanide complexes.^[22] Bands corresponding to Fe^{III} centers^[17,23] were not observed, indicating that the potassium iron hexacyanoferrate contains only divalent iron. The Raman spectrum of the potassium iron hexacyanoferrate electrode material also revealed the presence of water. As shown in Figure S5 in the Supporting Information, the band at 1640 cm^{-1} can be assigned to the bending vibration of hydrogen-bonded H_2O molecules.^[24] There are two strong bands at 2130 and 2091 cm^{-1} , which are assigned to the stretching vibration of carbon nitrogen triple bond.^[25] The bands at around 264 and 534 cm^{-1} correspond to stretching and bending modes of $\text{Fe}-\text{C}\equiv\text{N}-\text{Fe}$.^[26] Therefore, based on the crystal structure refinement, the formula of the potassium iron hexacyanoferrate is $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]\cdot 2\text{H}_2\text{O}$.

The morphology and detailed crystal structure of $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]\cdot 2\text{H}_2\text{O}$ were further analyzed by transmission electron microscopy (TEM) and selected area electron diffraction

(SAED) (Figure 2). TEM image (Figure 2a) shows uniform, well-packed cubes with edge lengths of $\approx 200\text{ nm}$. The SAED patterns (Figure 2c) of an individual cube (Figure 2b) can be indexed to the (020), (002), (011), and (011) crystal planes along [100] zone axis. In magnified TEM images (Figure 2d,e), the lattice of crystal planes ((011) and (011) with a d-spacing of $5.08\text{ \AA}$) can be observed, and the fast Fourier transform (FFT) patterns (inset in Figure 2d) are consistent with the SAED patterns. The atomically resolved high-angle annular bright-field scanning TEM (HAABF-STEM) image of the^[27] surface (Figure 2f) clearly shows the spaces within the $-\text{Fe}^{\text{II}}-\text{C}\equiv\text{N}-\text{Fe}^{\text{II}}-$ framework. The simulated atomic arrangement of the crystal planes^[27] (insets in Figure 2f,g) shows the square arrangement of atoms with interstitial sites along the [100] direction, which makes the as-prepared $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]\cdot 2\text{H}_2\text{O}$ large enough to accommodate potassium ions. Other TEM images (Figure S6, Supporting Information) show a uniform particle size distribution, along with morphological surface and crystal structural data. From the data presented above, the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]\cdot 2\text{H}_2\text{O}$ nanocubes present large interstitial sites and a 3D channel for

fast transport of cations. The uniform nanocube morphology with exposed crystal planes of (100) provides channels for the facile intercalation of potassium ions.

Electrochemical properties in aqueous electrolyte were characterized using a three-electrode system with 0.5 M K_2SO_4 as electrolyte, Ag/AgCl as a reference electrode, and a large piece of graphite as a counter electrode. Cyclic voltammetry (CV) measurements provide an effective method to investigate the electrochemical kinetics^[28] of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ nanocubes toward potassium-ion storage in aqueous electrolyte. Figure S7 in the Supporting Information shows CV data that feature two pairs of well-separated redox processes at +0.96/+0.88 V and +0.34/+0.23 V. At higher scanning rates, the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ electrodes present the same redox processes and well-defined reversibility (Figure S8, Supporting Information). Two redox processes are ascribed to the high-spin-state nitrogen-coordinated P site $\text{Fe}^{\text{III}}/\text{Fe}^{\text{II}}-\text{N}\equiv$ couple and low-spin-state carbon-coordinated R sites $\text{Fe}^{\text{III}}/\text{Fe}^{\text{II}}-\text{C}\equiv$ couple, respectively,^[29] according to Equation (1):



Accordingly, from the analysis of the voltammetric sweep data, the amount of charge stored by capacitive and intercalation processes can be distinguished, which can help gain further insight into the electrochemistry of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ nanocubes. Therefore, CV measurements at different scan rates (from 0.2 to 50 mV s^{-1}) were conducted as shown in Figure 3a. Both the cathodic and anodic current peaks increase with increasing scan rates while the potentials are almost invariant, confirming the low polarization of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ material in the aqueous electrolyte and the fast kinetics of the electrode when cycled under steady-state conditions.^[30] Analysis of the data at various sweep rates^[31] using the equation:

$$i = av^b \quad (2)$$

where i is current, v is scan rate, also provides insights into the potassium-ion insertion mechanism. When $b = 0.5$, a Faradaic intercalation process is indicated^[32] while a capacitive response is indicated if $b = 1$.^[31] A linear i versus v^b , where $b \approx 0.5$ (Figure 3c), displays a reversible diffusion-controlled potassium-ion insertion/extraction process (Table S1–S4, Supporting Information). The slopes of the corresponding $\log(v) - \log(i)$ plots of peak 1 and peak 4 in Figure 3a are 0.61 for both cathodic and anodic peaks at scan rates greater than 0.5 mV s^{-1} (Figure 3b), consistent with kinetics dominated by Faradaic intercalation processes.^[32] For the cathodic process below 0.5 mV s^{-1} , the slope decreases slightly to 0.57, consistent with a current that originates primarily from the potassium-ion intercalation process. The $\log(v) - \log(i)$ plots of the peak 2 and peak 3 in Figure 3a are shown in Figure S9 in the Supporting Information. The slopes of them are similar to peak 1 and peak 4, which are 0.59 for both cathodic and anodic processes, confirming that the kinetics were dominated by potassium-ion intercalation processes. The diffusion coefficient of potassium

ions in and out of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ electrode was further analyzed by the Randles–Sevcik equation:^[30]

$$i_p = 2.686 \times 10^5 n^{3/2} AD^{1/2} C v^{1/2} \quad (3)$$

where i_p is current maximum in amps, n is number of electrons transferred in the redox event (usually 1), A is electrode area in cm^2 , D is diffusion coefficient in $\text{cm}^2 \text{s}^{-1}$, C is potassium-ion concentration in mol cm^{-3} , v is scan rate in V s^{-1} . The diffusion coefficients for each oxidation and reduction process all are of the order of $10^{-5} \text{ cm}^2 \text{s}^{-1}$ for a range of scan rates (Figure S10, Supporting Information), which are three to four orders of magnitude greater than typical values for hexacyanoferrates in non-aqueous solutions.^[33] The diffusion coefficients for all processes increase with increasing scan rate, further confirming the high rate capability of the potassium-ion batteries developed in this work. The high diffusion coefficients are consistent with the moisture-dependent potassium-ion conductivity observed in Prussian Blue,^[33] where the aqueous environment facilitates ion diffusion^[34] which in turn facilitates access to the $\text{Fe}^{\text{III/II}}$ couples. Impedance spectroscopy measurements were used to investigate the physical processes that control the kinetic behavior of $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ in aqueous electrolyte (Figure S11, Supporting Information). A charge transfer resistance of 3 $\Omega \text{ cm}^{-2}$ was recorded, reflecting the high electron mobility in the cell.^[35,36]

The electrochemical performance of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ nanocubes was systematically investigated by galvanostatic charge and discharge measurements. As shown in Figure S12 in the Supporting Information, at 200 mA g^{-1} current density (1.4 C rate, fully charge or discharge the battery around 43 min), there are two charge/discharge plateaus at +0.92/+0.84 V and +0.3/+0.2 V, respectively, consistent with the CV results, which correspond to potassium-ion intercalation associated with two coordination sites. During the charge process (extraction of two potassium ions per formula unit) the electrode delivered a charge capacity of 140 mA h g^{-1} , consistent with its theoretical capacity for the two one-electron processes. The discharge capacity can reach 120 mA h g^{-1} and remain stable during the following cycles. After 100 cycles, there was a very small decline in capacity (115 mA h g^{-1} at discharge process), no shift of the charge/discharge plateaus and no change in the charge/discharge profiles were observed. Furthermore, the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ nanocubes exhibit remarkable rate performance and long-term cyclability as shown in Figure 3d,e, Figure S13 and S14 in the Supporting Information. Capacities of 111 mA h g^{-1} at current density of 500 mA g^{-1} (3.6 C rate), 109 mA h g^{-1} at current density of 1000 mA g^{-1} (7.2 C rate), and 104 mA h g^{-1} at current density of 2000 mA g^{-1} (14.4 C rate) were recorded. Even at a very high rate of 3000 mA g^{-1} (21.4 C rate), a reversible capacity of 93 mA h g^{-1} was observed. Reversibility at these high current densities was maintained without significant change of the charge and discharge profiles up to 500 cycles (Figure S11 and S12, Supporting Information). Figure 3e shows the high reversibility over 500 cycles at current densities of 500, 2000, and 3000 mA g^{-1} . The electrode retains reversible capacities of 107, 92, and 84 mA h g^{-1} after 500 cycles, respectively, which are 96%, 86%, and 81% of their initial capacities. Figure S15 in the Supporting Information shows the rate

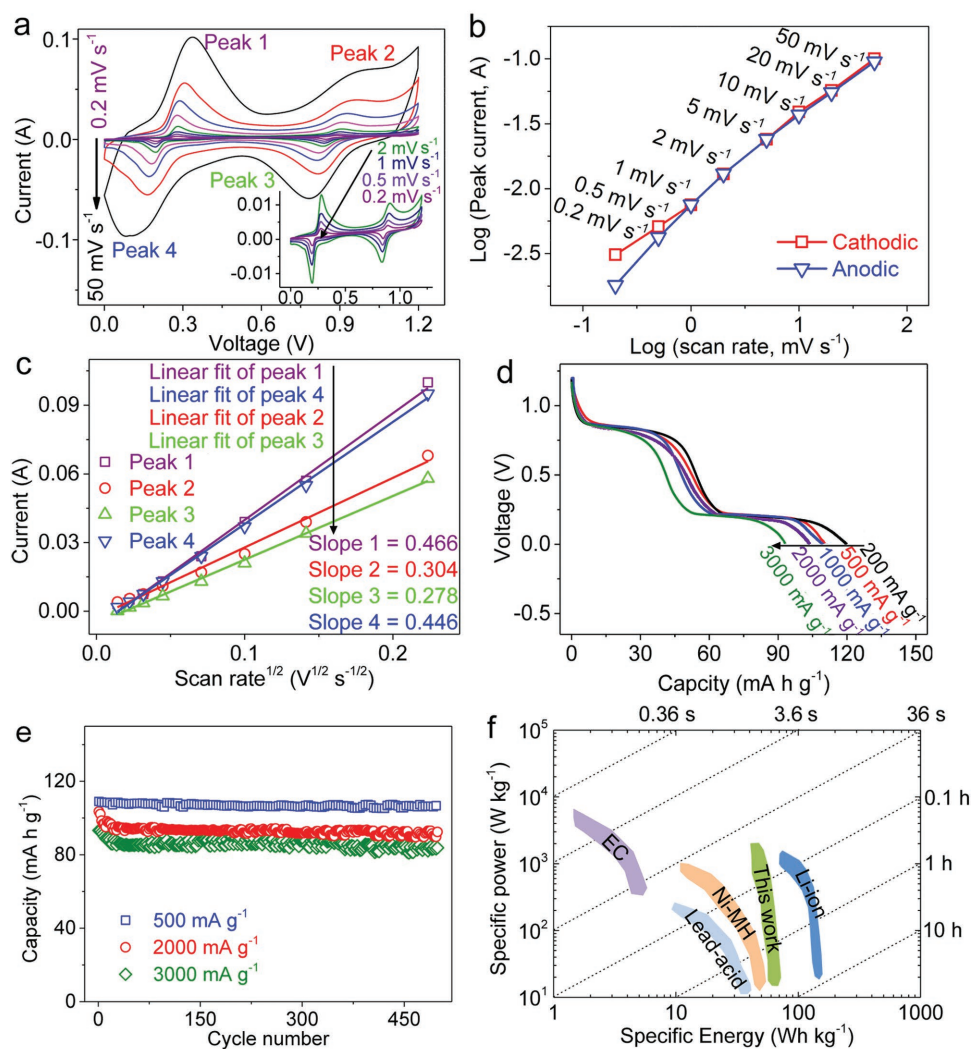


Figure 3. a) Cyclic voltammograms recorded at scan rates 0.2, 0.5, 1, 2, 5, 10, 20, and 50 mV s^{-1} . b) Log plot of peak 1 and peak 4 currents in a) versus scan rates. c) Plot showing peak current, i_p versus the square root of the scan rate, $\nu^{1/2}$. d) The first cycle profiles at current densities of 200, 500, 1000, 2000, and 3000 mA g^{-1} . e) Charge and discharge capacities versus cycle number at current densities of 500, 2000, and 3000 mA g^{-1} . f) Specific energy power for $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ based aqueous potassium-ion batteries (this work), as well as several other devices: electrochemical capacitors (EC),^[42] lead-acid batteries,^[38] nickel metal-hydride batteries,^[39] and lithium-ion batteries.^[43]

performance at various current densities. From which it can be seen that at low current density, high discharge capacity was obtained. Although the capacity decreased with increasing current densities, relatively high capacities were obtained (97 and 88 mA h g^{-1} at 2000 and 3000 mA h g^{-1} , respectively, 14.2 C and 21.4 C rates). A capacity of 118 mA h g^{-1} was recovered when the current density was returned to 200 mA g^{-1} (1.4 C rate). Moreover, the charging and discharging potential hysteresis observed at both low and high plateaus (Figure S16, Supporting Information) did not increase linearly with the current density (Figure S17, Supporting Information). Even at high current densities (up to 21.4 C), the difference between charge and discharge potentials was low. This is different from typical lithiation electrodes,^[37] where potential differences increase linearly with current density (Figure S17, Supporting Information). The small and stable voltage hysteresis of the

$\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ confirms that cells can be cycled with high round-trip energy efficiency at high cycling rates without the resistance limitation of the electrolyte.^[12b] Furthermore, the reversible capacity is significantly greater than other reported PBA-based cells (a comparison of capacities of different electrode materials for aqueous potassium-ion batteries can be found in Table S5, Supporting Information).

In contrast to the potassium iron hexacyanoferrate dihydrate electrode, an analogous nickel electrode ($\text{K}_2\text{Ni}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$, Figure S18–S21, Supporting Information) delivered less than 50 mA h g^{-1} (Figure S22 and 23, Supporting Information). The high oxidation state of the nickel metal ions in the large interstitial sites limits the electrochemical process to a one-electron reaction (Figure S24, Supporting Information). Furthermore, DFT calculations show that the high-spin Fe^{II} occupied nitrogen-coordinated

P sites facilitate faster potassium-ion diffusion compared with the Ni^{II} . As shown in Figure S25a in the Supporting Information, the migration energy barrier for potassium-ion diffusion within the lattice of $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ along the [100] direction (Figure S25c, Supporting Information, from I to III) is 1.66 eV, which is smaller than that of potassium-ion diffusion within the lattice of $\text{K}_2\text{Ni}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]$ (1.85 eV, Figure S26a, Supporting Information, from I to III). The energy barrier for potassium ions leaving the lattice of $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]$ is 4.3 eV (Figure S23d, Supporting Information, from I to III), which is much smaller than the energy barrier for potassium ion leaving the lattice of $\text{K}_2\text{Ni}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]$ (7.62 eV, Figure S26b, Supporting Information, from I to III). The activation energies are similar for $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ and $\text{K}_2\text{Ni}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]$ (1.0 and 1.1 eV, respectively, Figure S25a, Supporting Information). Figure S25b in the Supporting Information shows clear differences for the potassium ion leaving activation energies of $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ and $\text{K}_2\text{Ni}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]$ (0.97 and 4.02 eV, respectively). The migration energy barrier for potassium-ion diffusion within the lattice of $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ along the [100] direction is consistently lower than that of $\text{K}_2\text{Ni}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$.

As demonstrated here, the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ battery has a specific energy of $\approx 65 \text{ W h kg}^{-1}$ and a specific power of 1250 W kg^{-1} (Figure 3f). These properties are notable in that high specific power can be simultaneously achieved along with high specific energy, thus making these batteries potentially competitive against batteries such as lead-acid^[38] and nickel metal-hydride batteries for large-scale energy storage.^[39] Furthermore, the average operating potential of the aqueous $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ battery is close to + 0.6 V (vs Ag/AgCl), which is higher than those of common commercial MnO_2 and $\text{Ni}(\text{OH})_2$ batteries and the recently reported aqueous sodium-ion batteries,^[40] suggesting its promise for practical applications in large-scale energy storage.

To confirm the stability of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ cathode in the aqueous potassium-ion battery, we probed the changes in structure during potassium-ion extraction/insertion processes using ex situ XRD measurements as shown in Figure 4. All of the major diffraction peaks are maintained during the charge and discharge processes, indicating the stability of the potassium iron hexacyanoferrate dihydrate crystal structure. During the discharge process, the intensities of peaks corresponding to the (011) and (022) crystal planes in the initial orthorhombic phase gradually decreased. New peaks corresponding to the (200), (220), (202), and (400) crystal planes emerged during the early stages of ion extraction (charging). The (011), (200), and (400) peaks shifted monotonically toward higher 2θ angles upon further charging up to 1.2 V, suggesting potassium-ion extraction from the potassium iron hexacyanoferrate dihydrate phase occurs through a solid-solution reaction. The refined XRD patterns (Figure 4b,c) indicate the crystal structure of the charged electrodes maintains orthorhombic symmetry but with somewhat modified crystal parameters when half of the potassium ions are extracted from the lattice. When all potassium ions are extracted, shrinkage along the bc plane occurs accompanied by a slight expansion along the a axis. The corresponding refined structural models along a and c axes are shown in Figure S27a,b in the Supporting Information. The transitions

proceeded in the reverse manner when ions were inserted into the structure and the initial orthorhombic structure was restored at the end of discharge at 0 V with a slightly expanded crystal structure (Figure 4d,e, the refined structural models are shown in Figure S27c,d, Supporting Information). These results show that there is limited structural variation during the extraction/insertion process of potassium ions, which contributes to the superior cycling stability of the material. Moreover, ex situ FTIR spectra of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ electrodes were collected to probe the valence state change of iron during the first charge/discharge processes (Figure 4f). For $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ (region 1 in Figure 4f), the band in the region around 2060 cm^{-1} ^[23] can be fitted into three peaks at 2064, 2045, and 2002 cm^{-1} that match the divalent cyanide-coordinated Fe^{II} .^[17,22,23] Upon charging (region 2 in Figure 4f), a spectrum corresponding to that of trivalent ferricyanide was observed, with a new peak at 2142 cm^{-1} appearing.^[17,22,23] As more potassium-ions were extracted from the lattice, this new band grows in intensity (region 3 in Figure 4f). During the discharge process, the reverse trend was observed (regions 4 and 5 in Figure 4f). A Gatan parallel EELS with a high-resolution transmission electron microscopy (HRTEM) was used to acquire the Fe L_2 and L_3 spectra of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ electrodes at different charge and discharge states to confirm the valence state change of iron (Figure 4g). The Fe^{II} L_3 edge has a sharp maximum at $\approx 707.5 \text{ eV}$, whereas Fe^{3+} L_3 edges show maxima at ≈ 708.0 and 711.3 eV , consistent with reported results.^[41] Obviously, the bare electrode only presented the Fe^{II} (region 1 in Figure 4g). When the electrode was charged to 0.8 V, some Fe^{II} was oxidized to Fe^{III} , (region 2 in Figure 4g). When charging to 1.2 V, all Fe^{II} was oxidized to Fe^{III} (region 3 in Figure 4g). During the discharge process, Fe^{III} was reduced to Fe^{II} stepwise (regions 4 and 5 in Figure 4g). Furthermore, there was no morphology change during the potassium-ion insertion/extraction processes (Figure S28, Supporting Information). The structural stability of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ nanocubes was examined by collecting ex situ XRD patterns of a $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ electrode after 50 cycles at current density of 200 mA g^{-1} (Figure S29, Supporting Information). No crystal structure changes were evident when compared to the as-prepared $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ nanocubes. Ex situ SEM and TEM experiments also show that the morphology of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ electrode is unchanged after 50 cycles (Figure S30, Supporting Information). Figure S30a in the Supporting Information shows that the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ nanocube architecture is preserved after cycling. The corresponding SAED patterns (Figure S30b, Supporting Information), taken from the cycled electrode area in inset of Figure S30b in the Supporting Information can be indexed to the [100] zone axis, which indicates a well-defined crystallinity and structural stability after long cycling service.

The impressive electrochemical performance of the $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6] \cdot 2\text{H}_2\text{O}$ nanocube electrode material in aqueous potassium-ion batteries may be ascribed to the following factors: i) The high potassium-ion content and two Fe^{II} centers per formula unit, which can supply two electrons (per formula unit) and deliver exceptionally high capacities. ii) The uniform nanocube morphology with channels that allow facile intercalation and diffusion of potassium ions. iii) The open-framework

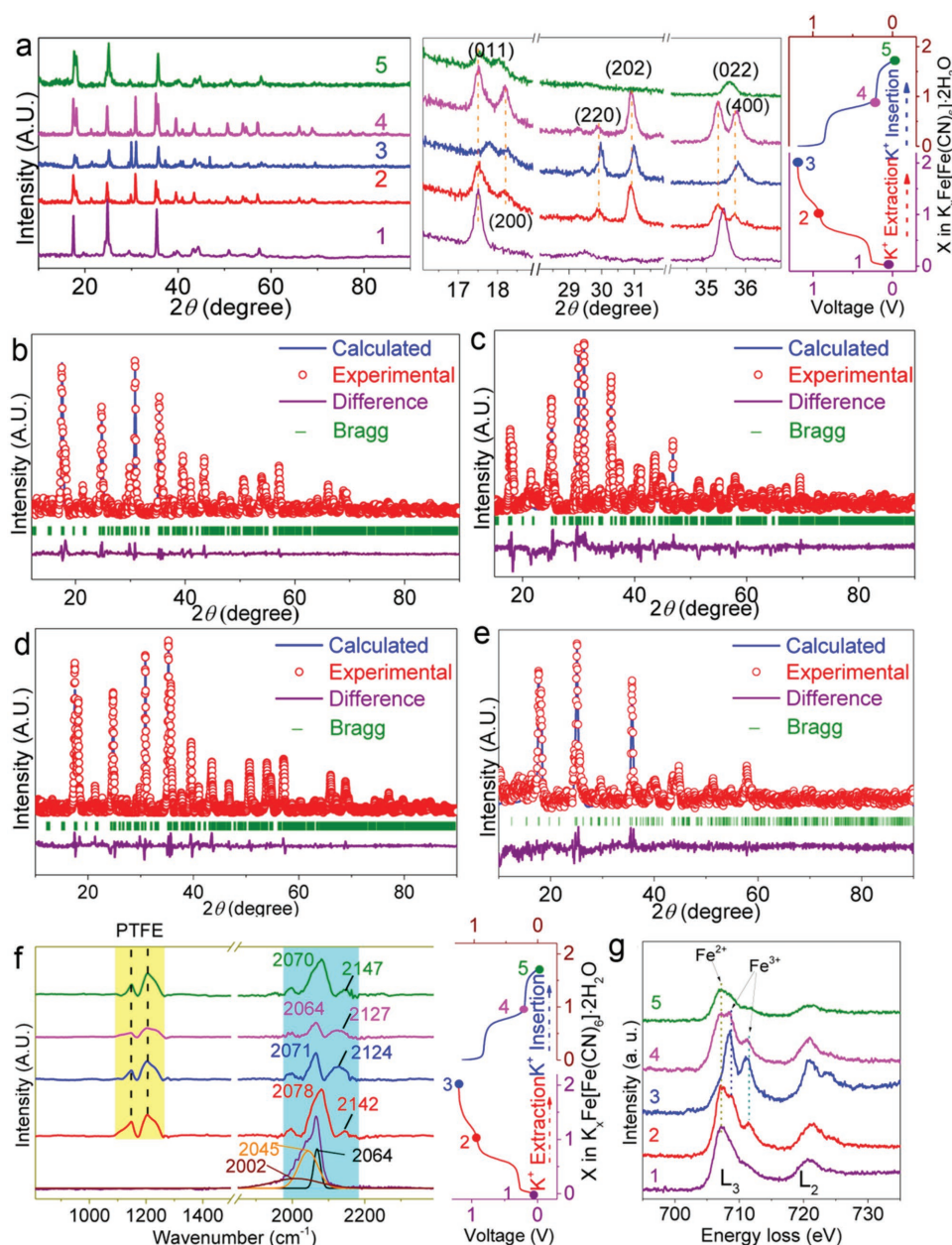


Figure 4. a) Ex situ XRD patterns of electrodes at different charge and discharge states as marked in the first galvanostatic charge and discharge curves of a $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]\cdot 2\text{H}_2\text{O}$ electrode (right column). Only the major diffraction peaks are shown. b–f) Rietveld refinement pattern of X-ray diffraction data of electrodes at different charge and discharge states as marked in (a). f) Ex situ FTIR spectra of a $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]\cdot 2\text{H}_2\text{O}$ electrode at different charge and discharge states (charge and discharge curves are shown in the right column). g) EELS spectra of iron cation in $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]\cdot 2\text{H}_2\text{O}$ electrode at different charge and discharge states.

structure that enables structural integrity while multiple potassium ions are accommodated, which leads to fast kinetics and highly reversible properties (Figure 3, and Figure S23 and S24, Supporting Information). iv) The interstitial water in the potassium iron hexacyanoferrate nanocubes that solvates ionic species in the $\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]$ framework, which in turn affects the mobility of potassium ions. v) The high ionic conductivity of the aqueous electrolyte, facilitating the fast intercalation and transport of potassium ions between the electrolyte and the lattice.

In summary, open-framework structured $\text{K}_2\text{Fe}^{\text{II}}[\text{Fe}^{\text{II}}(\text{CN})_6]\cdot 2\text{H}_2\text{O}$ nanocubes have been synthesized using ethylene glycol as a solvent by a low-temperature solvothermal method. The high potassium-ion content ensures that the new electrode material can function as an ample potassium-ion reservoir in aqueous potassium-ion batteries. A discharge capacity of 120 mA h g^{-1} (the highest value for an aqueous potassium-ion battery to date) was obtained with >85% capacity retention over 500 cycles at 21.4 C (to fully charge or discharge the batteries in $\approx 2.8 \text{ min}$). The high capacity can be ascribed to the high

potassium-ion content together with the two accessible one-electron redox processes. An open-framework structure enables structural integrity and facilitates the fast kinetics. Furthermore, all of the elemental components, in particular iron and potassium, are earth-abundant and nontoxic. These properties, together with the high capacity, outstanding cycle life, safety, and low cost of active materials make this battery become to the most attractive battery system for large-scale energy-storage applications.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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